

Exercise 4 - Counterproductive networks

Question 1

Generative Adversarial Network (GAN) (Simple) for generating MNIST-Fashion images

Dataset: MNIST-Fashion (28×28, 10 classes, black and white images).

Objective: Implement a basic GAN architecture and analyze the sensitivity of key hyperparameters on production quality.

(a) Simple GAN implementation

– Implement a simple GAN :

1 - Generator: Noise input with dimensions $z = 100$
 , Fully connected hidden layers, output 784 (28×28)

2 - Discriminator
 Input: 784 fully connected hidden layers, output 1 (real/fake)

– Use appropriate activation functions for hidden and output layers.

– Train the model for 100 training rounds with reasonable hyperparameters.

Produce and display 100. – At least 10 sample images in rounds of 20, 50

Analysis: Describe the dynamics of training and the development of visual quality.

(b) Learning rate sensitivity analysis

$\{1 \times 10^{-4}\}, 2 \times 10^{-4}$, – Experiment with different learning rates: $\{10^{-3} \times 1$

discriminator – Generator error and
 And monitor collapse mode indicators .

– Generate 10 sample images of the best checkpoint for each setting.

Analysis: Identify the optimal range of learning rates and explain the observed instabilities.

(c) Effect of noise dimensions

– Change the noise dimensions: {200 100, 32,} $\ddot{y} z$

- Keep other hyperparameters at optimal values from the previous analysis.

evaluate the diversity of the generated samples – Train each configuration and

- Use Score Inception or FID for quantitative comparison .

Analysis: Explain how noise dimensions affect the variety and quality of production.

and sustainability of education (d) Batch size

- Experiment with different batch sizes: {256, 128, 32,}

- Adjust the learning rate appropriately to maintain stability

- Record the speed of convergence and the final quality of the production

- Generate comparison networks including 10 samples for each batch size.

Analysis: Explain the trade-off between batch size, training stability, and computational efficiency.

Question 2

Advanced adversarial generative network architectures: DCGAN and CGAN

Conditional GAN for Fashion Generation

Objective: Implement and compare Convolutional Deep GAN architectures and

.MNIST

architecture analysis (A) DCGAN implementation and

- Implement a Deep Convolutional GAN :

Slow May 1 - Generator: Uses inverse convolutions to upsample noise into 28x28 images

Slow May 2 - Discriminator: Uses convolutional layers to classify images.

and appropriate organizing techniques. 3 - Includes BatchNorm

- Train with optimized hyperparameters from simple GAN analysis

- Experiment with different filter configurations: small [32, 64, 128] vs. large [128, 256, 512]

- Produce 10 sample images of the best model of each configuration

Analysis: Compare the visual quality and training stability of DCGAN with simple GAN and discuss the effect of filter capacity.

Do it.

and conditional generation (b) CGAN implementation

– Implement Conditional GAN by adding class conditioning to DCGAN :

1 - Generator: Takes both noise and class labels as input

2 - Discriminator: Takes both images and class labels as input

3 - Use embedding or one-hot encoding for class information.

– Train with optimized DCGAN hyperparameters

– Generate 10 samples for each of the 10 classes (total of 100 images)

Analysis: Evaluate the class-specific production quality and label fidelity compared to unconditional DCGAN .

(c) conditional production resistance (*Robustness*)

– Test the robustness of CGAN with noisy labels:

1 - Clean labels (0% noise)

2 - High noise (50% label displacement)

– Generate conditional-class examples under any conditions.

– Assess the degradation of production quality and class fidelity.

practical implications.

Analysis: Explain the sensitivity of CGAN to label quality and discuss the

Question 3

Comparative analysis and performance evaluation

Based on the results of questions 1-2 ^{CGAN} and **Objective:** Comprehensive performance comparison of Simple GAN,

DCGAN **Note:** This question only requires analysis of existing experimental results - no new experiments are required.

(A) Quantitative performance comparison

– Evaluate the best models from each architecture using the following criteria:

FID- 1

Inception Score - 2

3 - Training time and computational efficiency

4 - Using memory during training

- Create comprehensive comparison tables using the results of previous experiments

- Include 10 high-quality samples of each best model

trade-offs.

Analysis: Rank the architectures based on different evaluation criteria and explain the

(b) Visual quality assessment

- Accurate visual evaluation for specific fashion items (such as shirts, pants, shoes) in different architectures

- Identify common failure modes and artifacts for each approach

- Creating side-by-side comparison networks (10 samples for each architecture)

strengths and weaknesses of each architecture for producing fashion images. **Analysis:** Discuss the

(c) Analysis of the dynamics of education and sustainability

and compare convergence patterns across architectures. – Error curves

and analyze the recovery. – Occurrence of collapse mode

- Evaluate sensitivity to hyperparameter changes.

- Document training failures and successful configurations.

Analysis: Provide recommendations for stable GAN training on MNIST-Fashion .

(d) Computational efficiency and scalability

- Compare the training time per round and the total convergence time.

and analyze batch size limitations. – GPU memory requirements

- Evaluate the inference speed for generating samples.

- Discuss scalability to higher resolution or more complex datasets.

Analysis: Given computational constraints and quality requirements, suggest appropriate architectural choices.

Explanations

1 - All implementations must be done using a deep learning framework of your choice.

2 - Use the MNIST-Fashion dataset for all experiments.

3 - All generated images should be clearly labeled with architecture, hyperparameters, and training period.

and accurate analysis for each test.

4 - Reports should include error curves, sample networks produced

Implement Score Inception or use existing implementations.

and 5 - For quantitative assessment, FID

6 - Submit a compressed file named zip04.HW_StudentID containing the following:

– Complete implementation with clear documentation

– Checkpoints (trained model for best configurations)

and evaluation results – Produced images

– Comprehensive report in PDF format

7. Fraud or unauthorized collaboration is prohibited.

8 - The use of external resources or content generated by artificial intelligence for learning is permitted, but the work presented

It must be fully understood and explained by the student.

9 - For any questions about the practice, you can contact the teaching assistants via the following email:

ann.ceit.aut@gmail.com